

Quiz 3 - Answer Key

Thursday, March 5, 2026 8:16 AM

Question 1:

What is a page fault?

- ☐ When a process accesses a page that is not currently in physical memory
- ☐ When the CPU crashes due to memory corruption
- ☐ When a program exceeds its allocated memory space or access an invalid address
- ☐ When the OS swaps an entire process to disk

Question 2:

What is the primary function of a page table?

- ☐ To store all files in the system
- ☐ To map virtual addresses to physical addresses
- ☐ To track user permissions for accessing memory address space
- ☐ To manage process scheduling

Question 3:

Why can bitmaps be inefficient in some memory allocation scenarios?

- ☐ They use excessive pointer storage
- ☐ They cannot track small blocks
- ☐ Scanning for free space can be slow
- ☐ They always fragment memory

Question 4:

What is the purpose of the Translation Lookaside Buffer (TLB)?

- ☐ To translate virtual addresses to physical addresses more efficiently
- ☐ To store program instructions in the memory management unit (MMU)
- ☐ To handle segmentation faults for faster recovery
- ☐ To track free memory pages in a cache on the CPU

Question 5:

What is one key benefit of virtual memory in a modern operating system?

- ☐ It lets multiple processes share the same physical memory safely
- ☐ It ensures all pages remain resident in main memory
- ☐ It removes the need for page tables
- ☐ It allows the CPU to bypass the MMU during address translation

Question 6:

The Least Recently Used algorithm requires adding a buffer for each page table entry in the page table as well as a bitmap. Explain the role the (1) buffers and (2) bitmap play in the LRU demand paging system.

Answer:

Question 7:

Discuss the changes required to an operating system's (1) global variables, (2) process table entries, and (3) page fault handler to implement Page Fault Frequency prepaging system.

Answer: